CrowdWisdomTrading Macro economics agent Intern position assessment

The document gives a brief introduction to an assignment to check your qualifications for the internship as a Quantitative Data Scientist. Please complete it on your own time and submit it within 1 week.

Project Overview
Create a Python-based machine learning pipeline that predicts the optimal trading strategy parameters (permutations) for a future period. The system will ingest historical trading logs, scrape macroeconomic event calendars, join the datasets in a database, and train a predictive model using rigorous out-of-sample validation.

Technical Requirements
https://apify.com/ - for data scraping (alternative data features)

Project Scope
Develop macro economic prediction model

1. Macroeconomic Event Ingestion (Apify-based)
    - Scrape a macroeconomic calendar (CPI releases, FOMC announcements, employment reports) for the last 180 days using Apify.
    - Extract transaction timestamps (UTC/EST alignment), event names, actual values, and forecasted values.
    - most recent and relevant economic news (use tavily.com)
2. Quantitative Feature Engineering & Macro summary
    - Feature Engineering: Create time-based features
    - 1 page summary of the macro economic outlook
3. Machine Learning Model & Walk-Forward Validation
    - Build a predictive model to forecast expected trade P&L or win rate for each strategy permutation.
4. Visual Matrix Output Generator
    - Charts that can visual the macro sentiment change as time changes

Submission by email gilad@crowdwisdomtrading.com including apify/tavily/exa

Submit the following:

1. A link to your GitHub/GitLab repository.
2. APIFY /tavily/exa tokens and actor IDs used for scraping (so we can reproduce your pipeline).

3. An evaluation report (PDF or Markdown) detailing model performance metrics (Sharpe Ratio, Sortino Ratio, Max Drawdown).

Timeline

You will have 5 - 7 days to complete this task. Please ensure you submit the deliverables by the end of this period.

We look forward to seeing your solution\!

Thanks for applying!
Gilad
CrowdWisdomTrading CEO